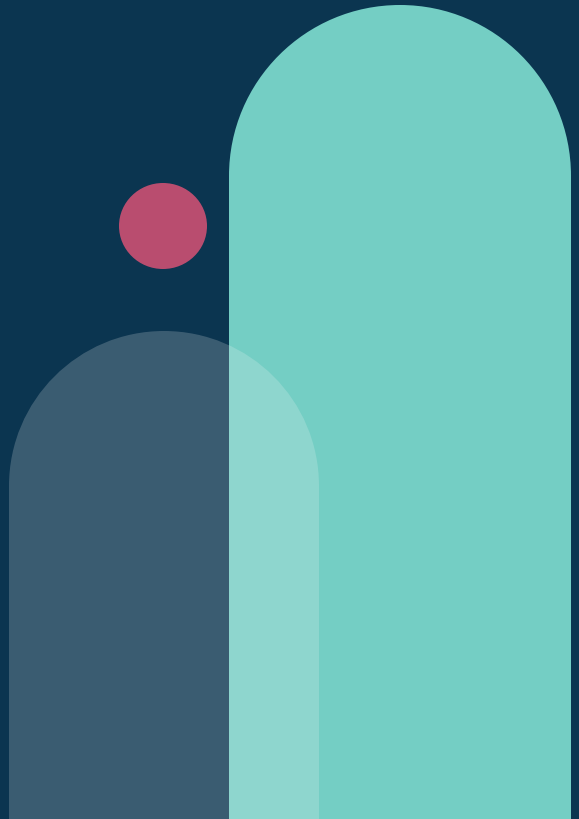
Basics of AI, ML, and DL

Introduction

01

What is Artificial Intelligence (AI)?



What is AI?

Definition of AI:

AI refers to the simulation of human intelligence in machines that are designed to think and act like humans.

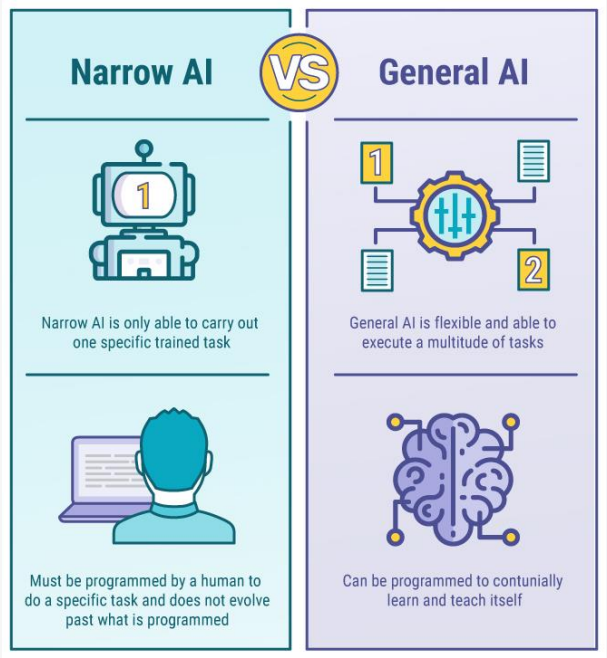
Types of AI:

Narrow AI (Weak AI): Designed for a specific task, like price forecaster or image recognition software.

General AI (Strong AI): A hypothetical type of AI that possesses the ability to perform any intellectual task that a human can do.

Real-world examples of AI applications:

- **Voice Assistants:** Siri and Alexa
- **Recommendation Systems:** Netflix and Youtube
- **Autonomous Vehicles:** Tesla, Genesis, BMW



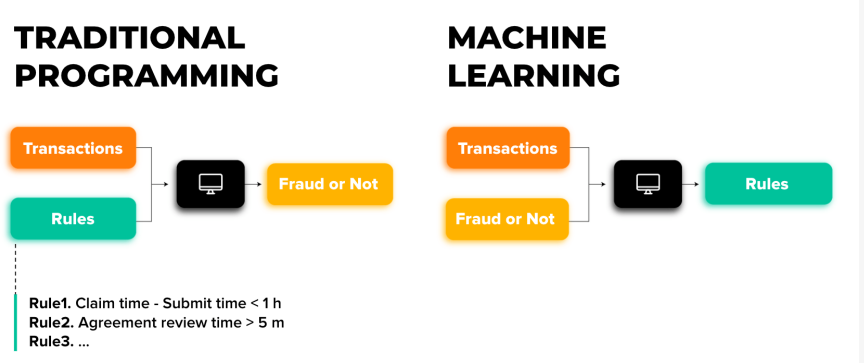
What is ML?

Definition of Machine Learning:

Machine Learning is a subset of AI that focuses on the development of algorithms that can learn from and make decisions based on data.

How ML differs from traditional programming:

- **Traditional Programming:** A programmer writes explicit rules for the computer to follow. For example, a spam filter
- **Machine Learning:** The machine then learns to identify patterns in the data, a large set of labelled emails (some marked as spam, others not), and can automatically classify new emails as spam or not.



Key use cases of ML in various industries:

- **Healthcare:** Detecting cancer in medical images
- **Finance:** Credit scoring, fraud detection
- **Retail:** E-commerce platforms

What is DL?

Definition of Deep Learning:

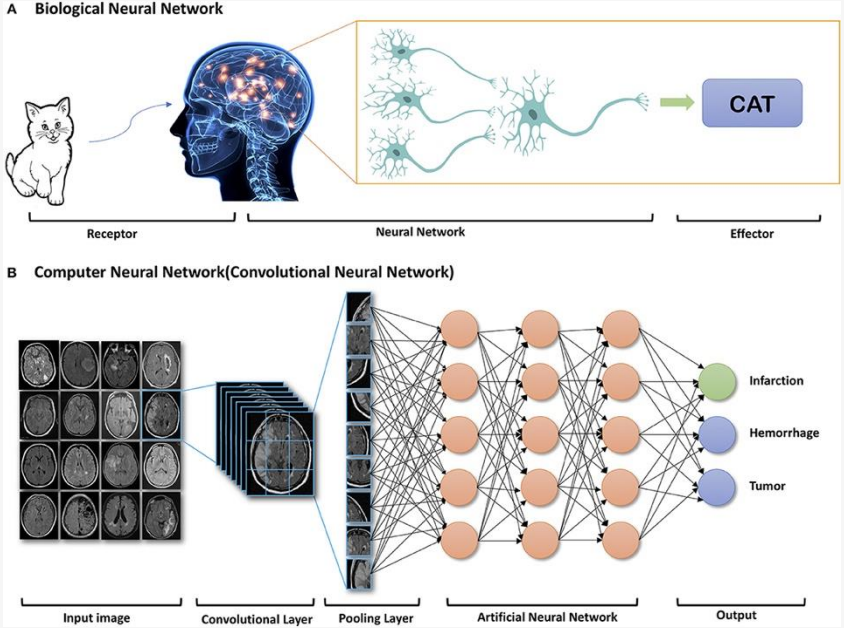
Deep Learning is a subset of Machine Learning that uses neural networks with many layers (hence "deep") to model complex patterns in large datasets.

Introduction to Neural Networks:

- **Neural Networks:** Inspired by the structure of the human brain (neurons).
- **Layers in Neural Networks:** Typically, neural networks have an input layer, multiple hidden layers, and an output layer.

Key applications of DL:

- **Image Recognition:** Facial recognition systems,
- **Natural Language Processing (NLP):** Google Translate, sentiment analysis, and chatbots.
- **Autonomous Vehicles:** DL models process sensor data from cameras, radar, and LiDAR to identify objects and make driving decisions.



Relationship between AI, ML, and DL

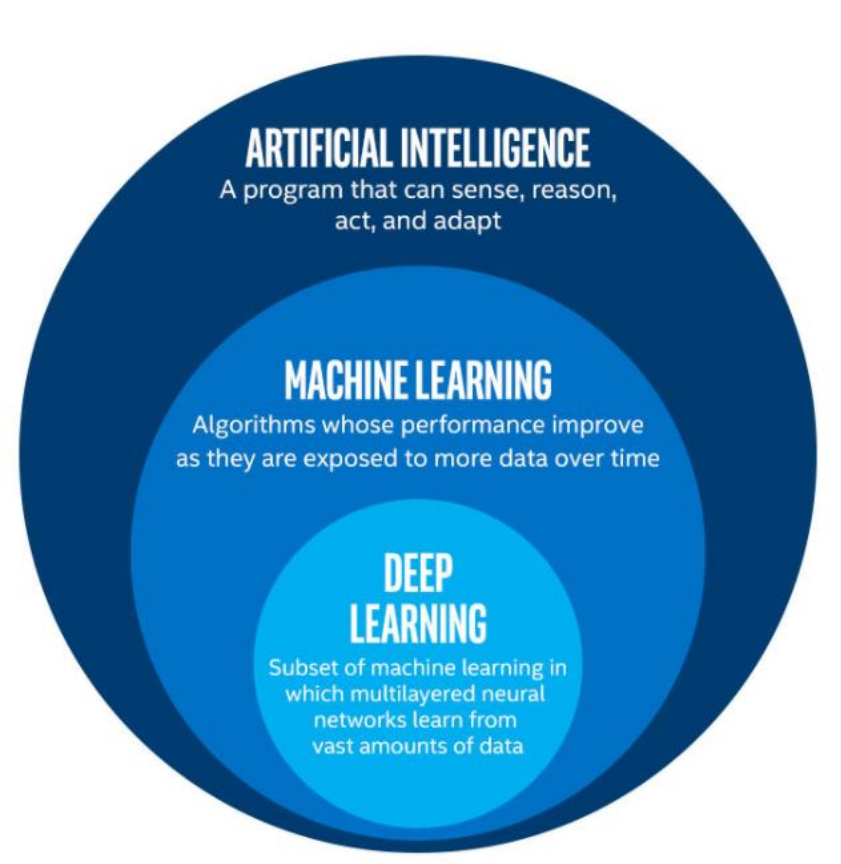
AI: The broadest field, including any technique that enables machines to **mimic** human intelligence.

ML: A subset of AI that allows machines to **learn from data** and improve over time without being explicitly programmed.

DL: A subset of ML that involves neural networks with many layers, capable of learning from vast amounts of data.

The importance of each in modern technology:

- **AI:** The driving force behind intelligent systems, from recommendation engines to complex decision-making processes.
- **ML:** Essential for analyzing data and making predictions across industries.
- **DL:** Critical for tasks requiring high accuracy, such as image recognition, speech synthesis, and autonomous driving.



02

Historical Context and Evolution of AI

History and Evolution

The Beginnings:

- **1950:** Alan Turing proposes the Turing Test for machine intelligence.
- **1956:** John McCarthy coins "Artificial Intelligence" at the Dartmouth Conference.

Early AI Programs:

- **1950s-1960s:** Logic Theorist and ELIZA demonstrate early AI capabilities.
- **1970s-1980s:** Expert systems like MYCIN mimic human decision-making.

AI Winters:

- **1970s-1990s:** AI funding declines due to unmet expectations.

Rise of ML and DL:

- **1980s-1990s:** ML gains traction with algorithms like Decision Trees and SVMs.
- **1986:** Backpropagation popularized for training neural networks.
- **2006:** Deep Learning re-emerges with Hinton’s Deep Belief Networks.
- **2012:** Hinton’s team wins ImageNet with deep neural networks.

Modern Era of AI:

- **2010s:** Advances in GPUs, big data, and algorithms drive AI breakthroughs.
- **2018-Present:** Transformers revolutionize NLP with models like BERT and GPT-3.

Current Trends in AI

Generative AI

- Automates content creation across text, images, video, and code.
- Democratizes AI, making it accessible to non-technical users.

Multimodal AI

- Combines text, audio, video, and images for richer interactions.
- Enhances search, content creation, and decision-making processes.

Small Language Models (SLMs)

- Offer high performance with fewer computational resources.
- Enable AI implementation in mobile and hardware-constrained devices.
- Support versatile robotics for general-purpose applications.

AI for Workplace Productivity

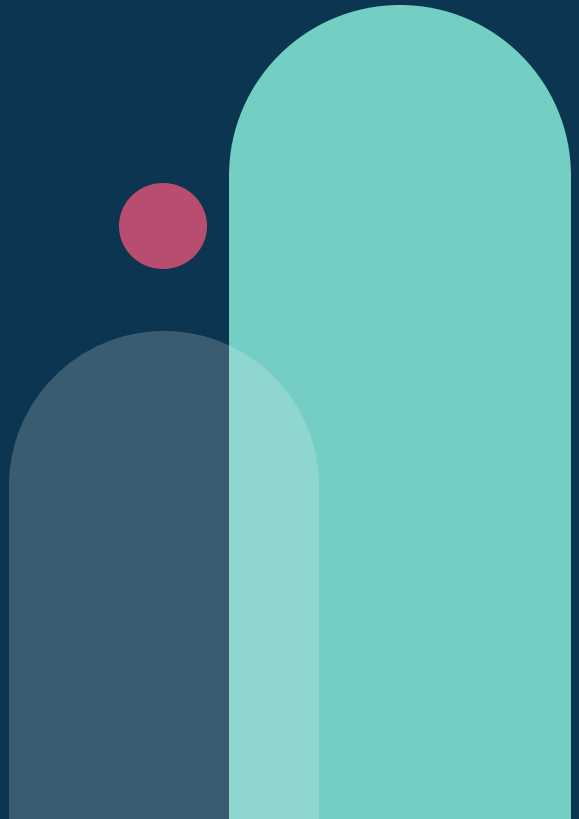
- Automates repetitive tasks, boosting human productivity.
- Allows focus on creativity, emotional intelligence, and moral judgment.
- Integrates AI directly into interfaces for enhanced efficiency.

New Use Cases Across Industries

- Extends AI's impact to education, healthcare, manufacturing, and more.
- Enables innovative solutions, scalability, accuracy, and faster task completion.

03

Concepts in ML



Understanding Data, Features, and Labels

What is Data in ML?

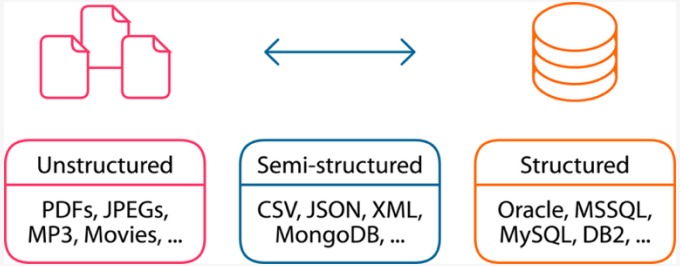
- Data is the raw input that ML models learn from, available in structured and unstructured forms.

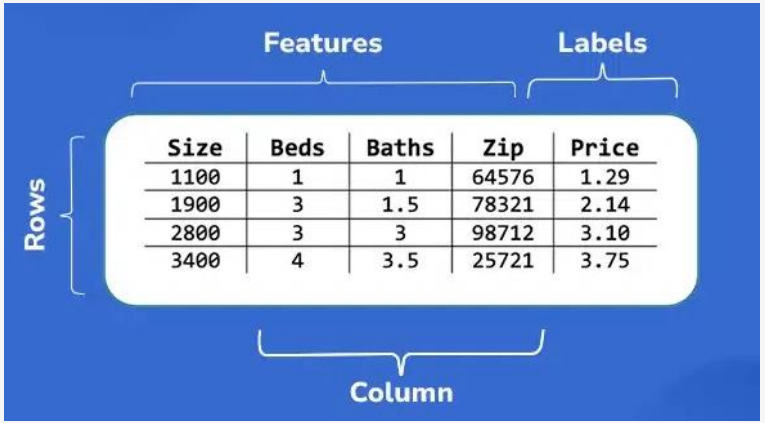
Features:

- Features are individual measurable properties or characteristics of the data, used as input to ML models.
- Example: In a dataset of houses, features could include square footage, number of bedrooms, and location.

Labels:

- Labels are the target outcomes or variables that the model is trained to predict, typically used in supervised learning.
- Example: In the house dataset, the label could be the house price.





Size	Beds	Baths	Zip	Price
1100	1	1	64576	1.29
1900	3	1.5	78321	2.14
2800	3	3	98712	3.10
3400	4	3.5	25721	3.75

Supervised Learning

Definition:

The model is trained on labelled data, learning to predict outputs from input features.

Examples:

Classification:

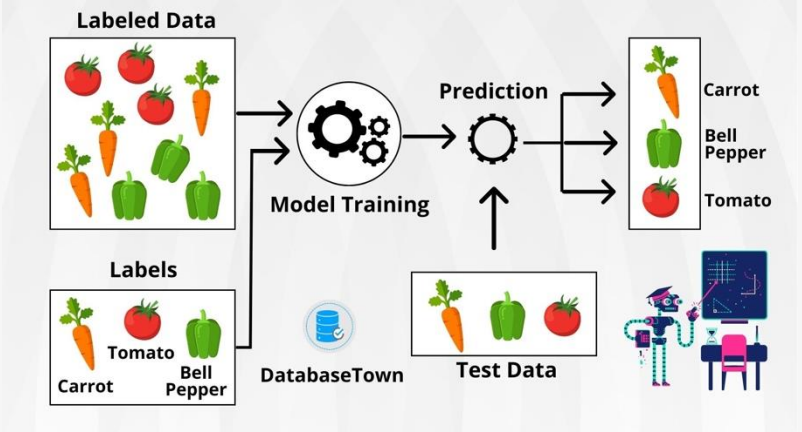
- Predicting whether an email is spam or not.
- Identifying if a given image contains a cat or not.

Regression:

- Predicting house prices based on features like size, location.
- Estimating a person's age based on their physical activity data.

Applications:

Medical diagnosis, fraud detection, stock price prediction.



Unsupervised Learning

Definition:

The model is trained on unlabeled data, identifying patterns, structures, or groups within the data.

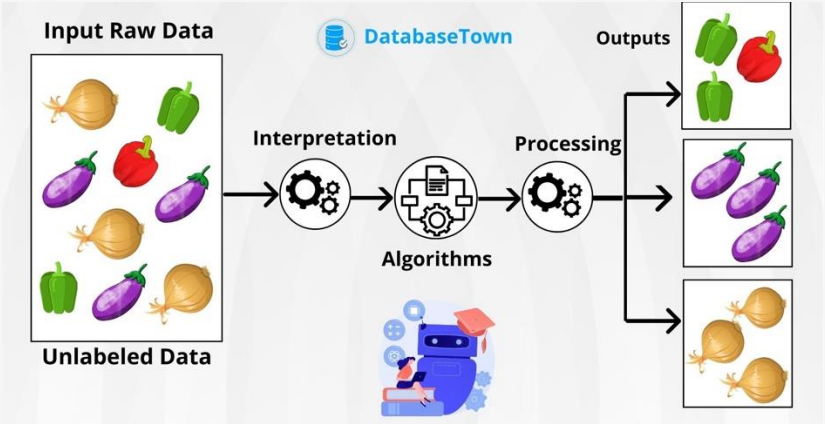
Examples:

Clustering:

- Grouping customers based on purchasing behavior.
- Segmentation of similar documents in a large corpus.

Dimensionality Reduction:

- Reducing the number of variables in a dataset (e.g., PCA).
- Visualizing complex datasets in 2D or 3D.



Applications:

Market segmentation, anomaly detection, recommendation systems.

Training, Testing, and Validation

Training:

The model learns from the training dataset, adjusting parameters to minimize errors.

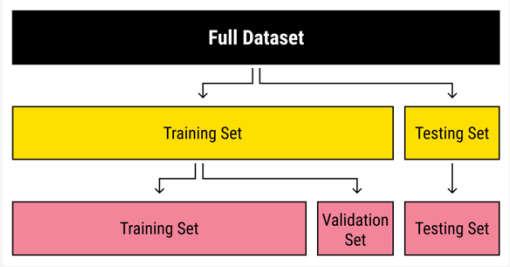
Testing:

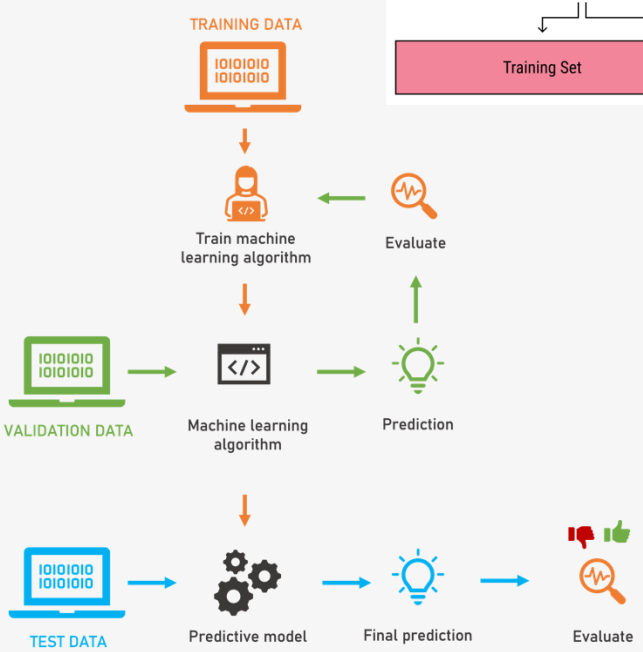
The model's performance is evaluated on unseen data (test set) to measure generalization ability.

Validation:

The model's hyperparameters are fine-tuned using a validation set, helping prevent overfitting.

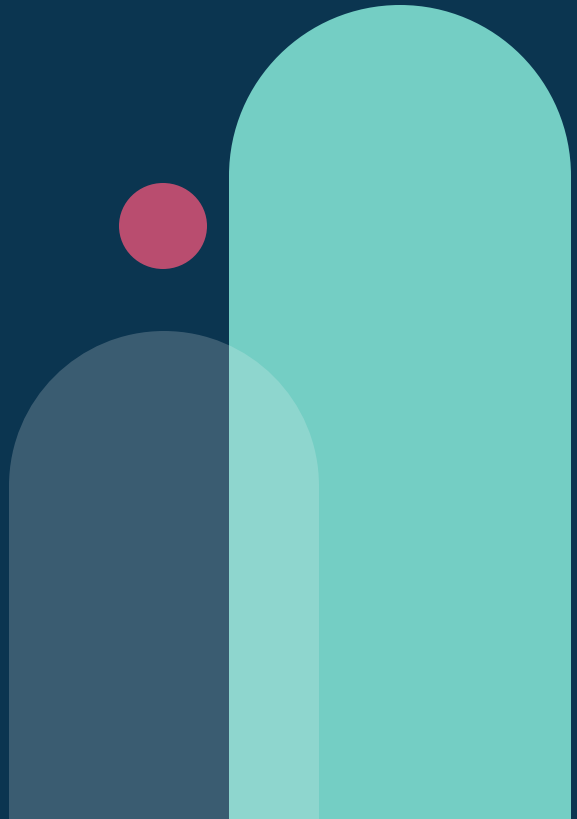
Common techniques: Cross-validation, grid search.





04

Overfitting vs Underfitting



What is Overfitting?

Definition:

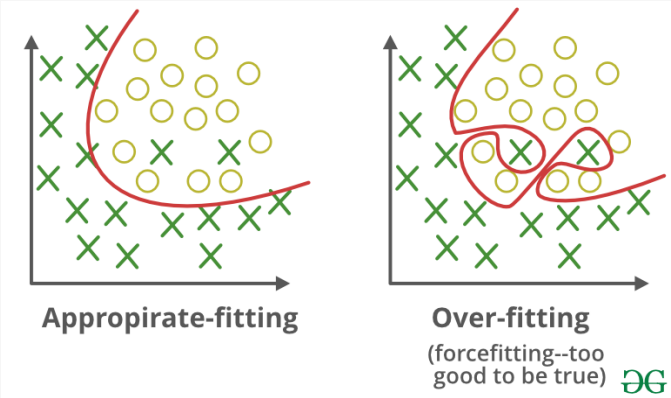
- Overfitting occurs when a model learns the training data too well, capturing noise and details that don't generalize to new data.

Causes:

- Model complexity is too high (e.g., too many parameters).
- Training on a small dataset with a lot of variance.

Prevention Techniques:

- **Regularization:** Adding a penalty to the model for having too many parameters.
- **Cross-Validation:** Splitting the data into multiple subsets to validate model performance.
- **Early Stopping:** Halting training when performance on a validation set starts to degrade



What is Underfitting?

Definition:

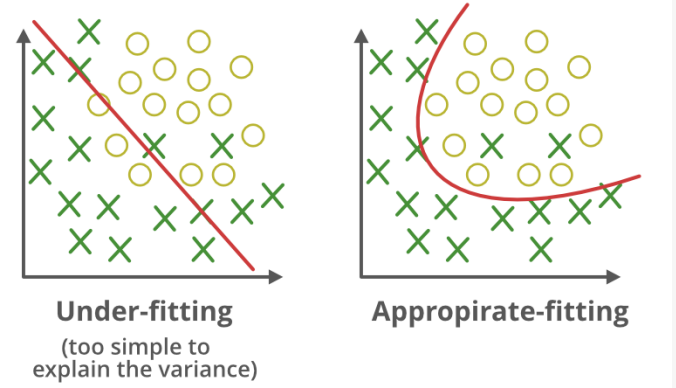
- Underfitting occurs when a model is too simple to capture the underlying patterns in the data, leading to poor performance on both training and test data.

Causes:

- Model complexity is too low (e.g., not enough parameters or features).
- Insufficient training time or poor data quality.

Solutions:

- **Increase Model Complexity:** Add more features or use a more sophisticated model.
- **More Data:** Providing additional training data can help the model learn better patterns.
- **Feature Engineering:** Create new features that better capture the underlying patterns.



Overfitting vs Underfitting

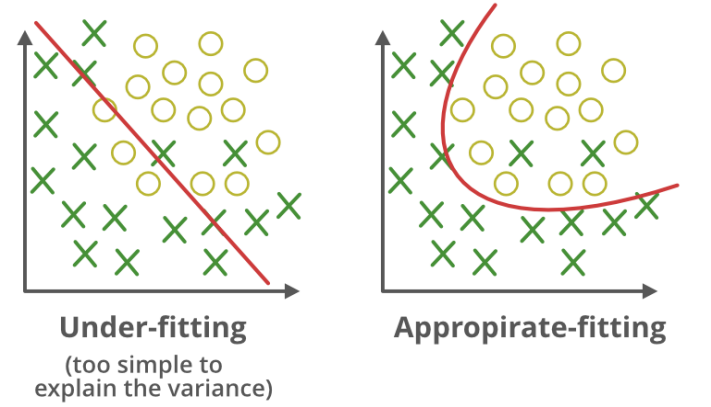
Key Differences:

- **Overfitting:** Model captures noise; high variance and low bias.

  Student who memorizes answers for a test but fails when the questions change slightly.

- **Underfitting:** Model misses patterns; high bias and low variance.

  Student who only reads the summaries and misses key details.



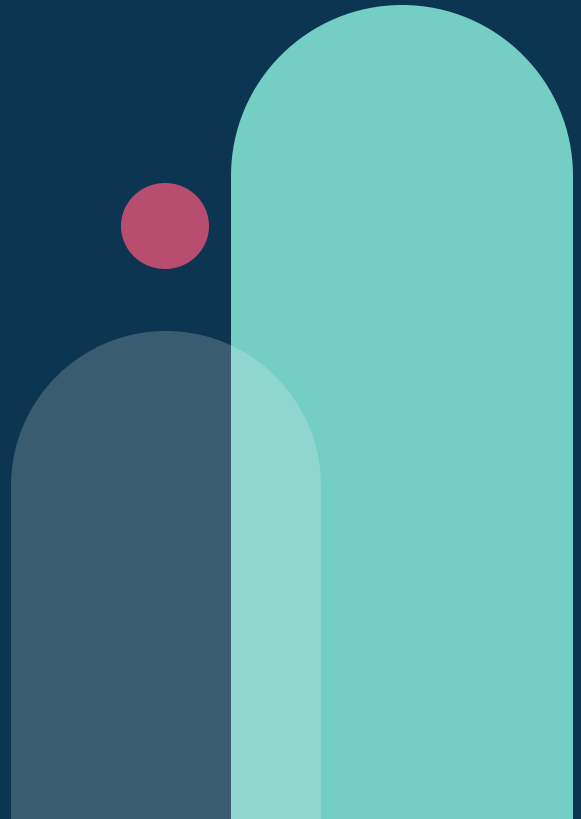
Balancing the Trade-Off:

The goal is to find the sweet spot where the model generalizes well, avoiding both overfitting and underfitting.

not too simple, not too complex—just right

05

scikit-learn

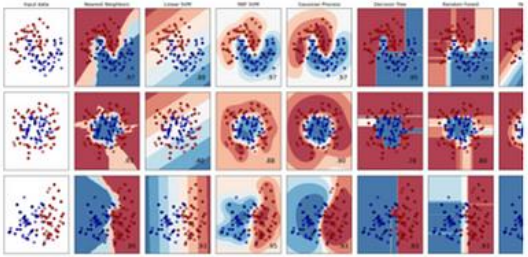


Why scikit-learn?

Classification

Identifying which category an object belongs to.

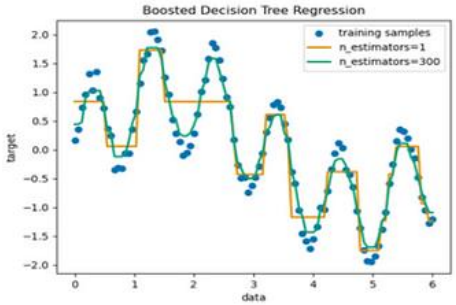
Applications: Spam detection, image recognition.
Algorithms: SVM, nearest neighbors, random forest, and more...



Regression

Predicting a continuous-valued attribute associated with an object.

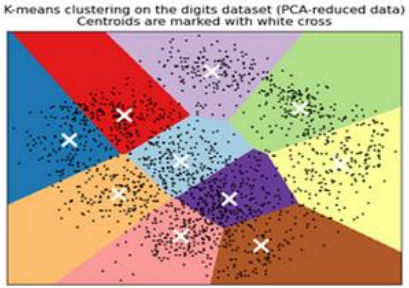
Applications: Drug response, Stock prices.
Algorithms: SVR, nearest neighbors, random forest, and more...



Clustering

Automatic grouping of similar objects into sets.

Applications: Customer segmentation, Grouping experiment outcomes
Algorithms: k-Means, spectral clustering, mean-shift, and more...



What is scikit-learn?

What is scikit-learn?

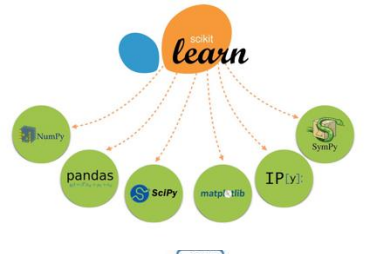
- An open-source Python library for machine learning.
- Provides simple and efficient tools for data mining and data analysis.

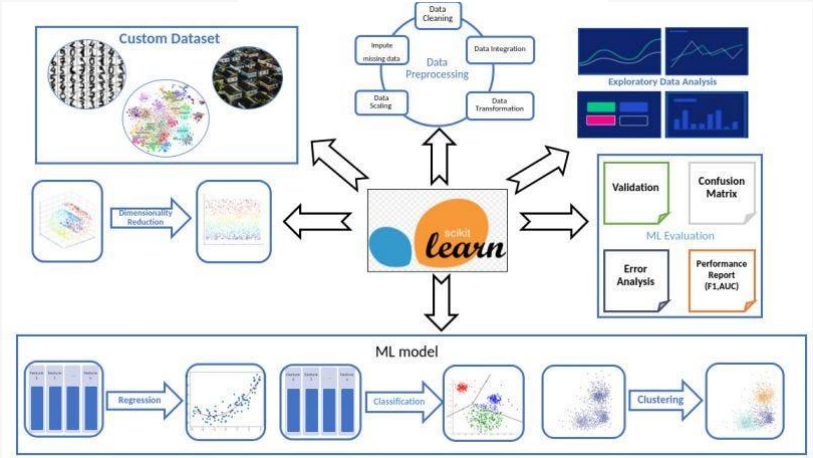
Key Features:

- **Wide Range of Algorithms:** Supports classification, regression, clustering, and more.
- **Built-in Datasets:** Includes datasets like Iris, Wine, and Digits for practice.
- **Model Selection Tools:** Provides cross-validation, grid search, and metrics for evaluating models.

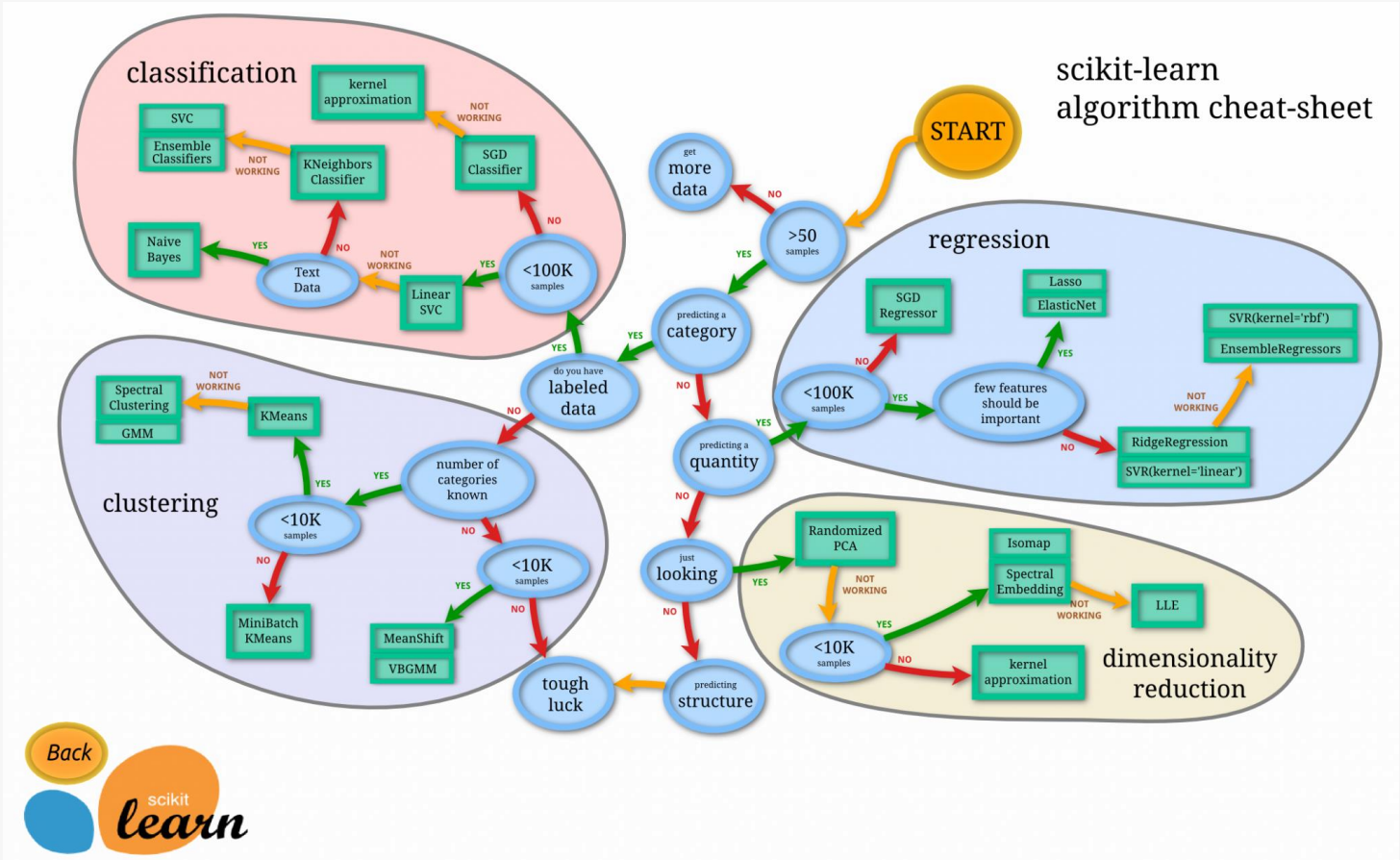
Why Use scikit-learn?

- **Ease of Use:** Simple API for implementing and testing models.
- **Integration:** Works seamlessly with other Python libraries like NumPy and Pandas.
- **Community Support:** Extensive documentation and active user community.





scikit-learn



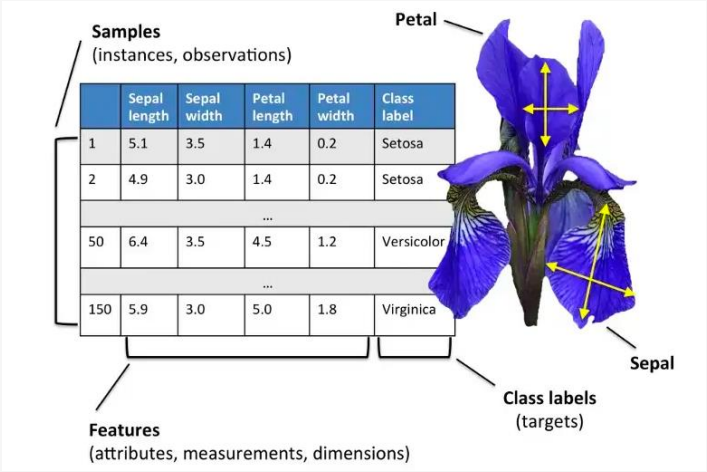
Classification Example

Problem Statement:

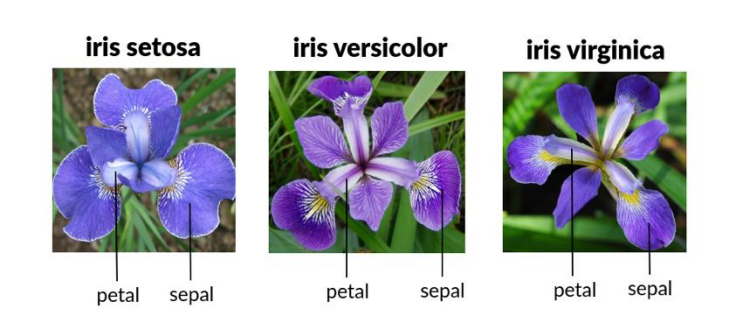
Classify iris flowers into three species (Setosa, Versicolor, Virginica) based on their features (sepal length, sepal width, petal length, petal width).

Explanation:

- **Logistic Regression:** A simple, interpretable model that's effective for small datasets like Iris.
- **Features:** Sepal length, sepal width, petal length, petal width are used as inputs to predict the flower species.
- **Output:** The model predicts the species of each flower and compares it with the actual species to evaluate accuracy.

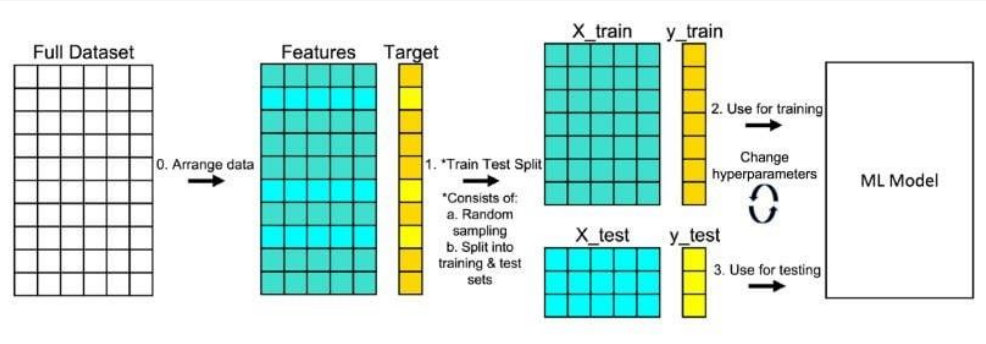


	Sepal length	Sepal width	Petal length	Petal width	Class label
1	5.1	3.5	1.4	0.2	Setosa
2	4.9	3.0	1.4	0.2	Setosa
...					
50	6.4	3.5	4.5	1.2	Versicolor
...					
150	5.9	3.0	5.0	1.8	Virginica



Classification Example

1. **Import Libraries:** Import scikit-learn and necessary modules.
2. **Load Dataset:** Use datasets.load_iris() to load the Iris dataset.
3. **Split Data:** Use train_test_split to create training and testing sets.
4. **Choose a Model:** Select a simple classifier, such as Logistic Regression.
5. **Train the Model:** Fit the model on the training data using model.fit().
6. **Predict and Evaluate:** Make predictions on the test set and evaluate accuracy using model.score().



Thanks!

Do you have any questions?

